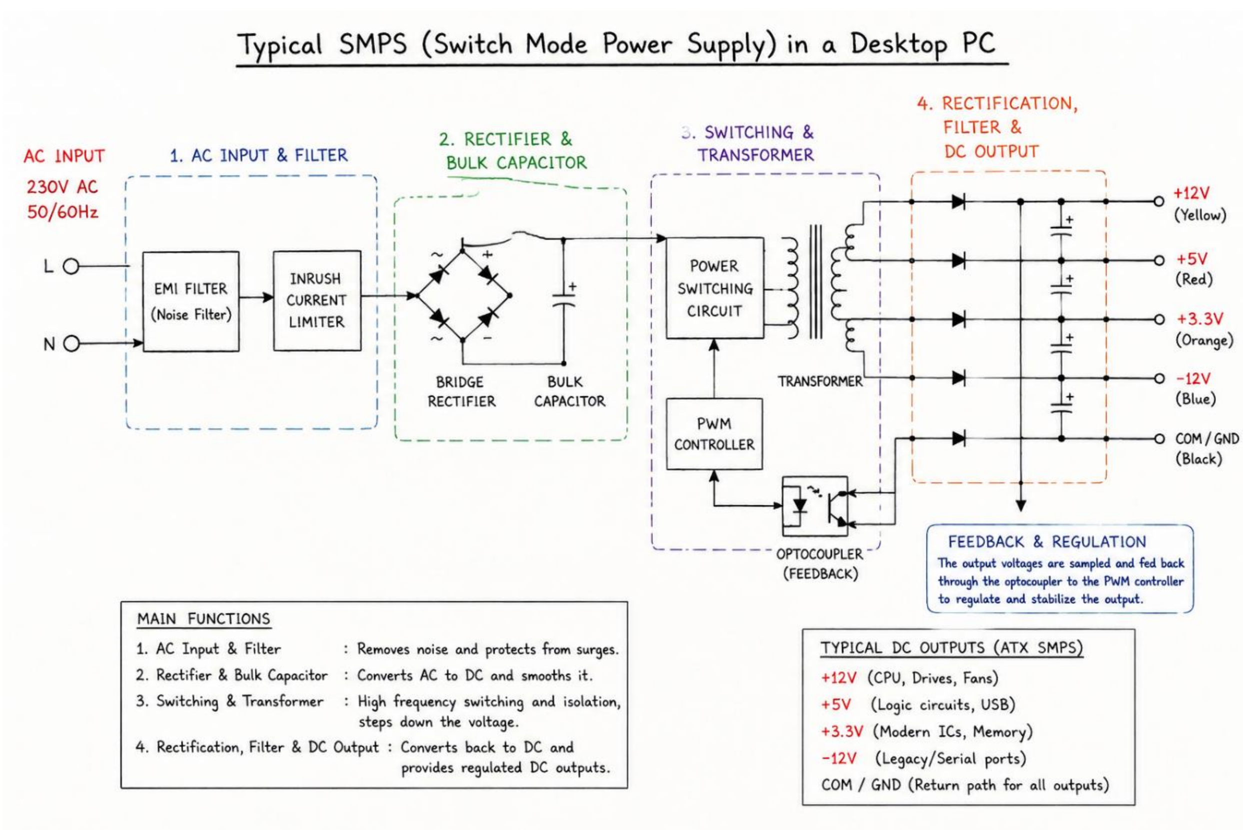
1.List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply

Device	Power Supply Type	Why It Needs This Power Supply
Desktop Computer	SMPS	Converts AC power to regulated DC power for computer components.
LCD/LED Monitor	SMPS	Provides efficient and stable power to the display circuits.
Laptop Charger	SMPS	Converts high-voltage AC power into low-voltage DC power for charging the laptop battery.
Desktop Computer (with UPS)	UPS	Keeps the computer running during power cuts and prevents data loss.
Wi-Fi Router (with UPS)	UPS	Maintains internet connectivity during power outages.

2.Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw\.io.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting or disconnecting.

Item	Details
Name	ATX 24-Pin Motherboard Connector
Powers	Motherboard, CPU power circuits, RAM slots, PCIe slots
Safety Tip	Turn off and unplug the PSU before connecting or disconnecting. Press the locking clip before removing it.

Item	Details
Name	SATA Power Connector
Powers	SSDs, HDDs, SATA DVD/CD drives
Safety Tip	Align the L-shaped connector correctly and never force it into the device.

Item	Details
Name	Molex 4-Pin Connector
Powers	Older HDDs, optical drives, cooling fans, accessories
Safety Tip	Pull the connector straight out; do not pull on the wires.

Item	Details
Name	PCIe 6-Pin / 8-Pin Power Connector
Powers	Graphics Cards (GPUs) and high-performance expansion cards
Safety Tip	Ensure the connector clicks into place completely before powering on the PC.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step	Component	Cable/Connector Used	Procedure
1	Motherboard	**24-Pin ATX Power Connector**	Connect the 24-pin ATX cable from the SMPS to the motherboard's main power socket until the locking clip clicks into place.
2	CPU	**8-Pin EPS (CPU Power) Connector**	Connect the 8-pin CPU power cable near the CPU socket on the motherboard. Some boards may use a 4+4 pin connector.
3	Graphics Card (GPU)	**PCIe 6-Pin or 8-Pin Connector**	Insert the graphics card into the PCIe slot and connect the required PCIe power cable(s) from the SMPS.
4	SSD/HDD	**SATA Power Connector**	Connect a SATA power cable from the SMPS to each SSD or HDD.
5	SATA DVD Drive (if used)	**SATA Power Connector**	Connect another SATA power connector to the optical drive.
6	Case Fans/Accessories	**Molex or SATA Power Connector**	Connect cooling fans, RGB hubs, or other accessories using the appropriate connector.
7	Cable Check	All Components	Verify that all power connectors are firmly seated and locked before turning on the PC.
8	Power On	Entire System	Connect the AC power cable to the SMPS, switch the PSU to "ON", and start the PC.